Contacts:

Michael Lin / David Wong
UMC, Investor Relations
+ 886-2-2658-9168, ext. 16900
jinhong_lin@umc.com
david_wong@umc.com

UMC Reports Fourth Quarter 2022 Results

Full-year 2022 operating income exceeds NT$100bn with automotive revenue up 82% YoY

Fourth Quarter 2022 Overview[1]:

- **Revenue: NT$67.84 billion (US$2.21 billion)**
- **Gross margin: 42.9%; Operating margin: 34.8%**
- **Revenue from 22/28nm: 28%**
- **Capacity utilization rate: 90%**
- **Net income attributable to shareholders of the parent: NT$19.1 billion (US$621 million)**
- **Earnings per share: NT$1.54; earnings per ADS: US$0.251**

Taipei, Taiwan, ROC – January 16, 2023 – United Microelectronics Corporation (NYSE: UMC; TWSE: 2303) ("UMC" or "The Company"), a leading global semiconductor foundry, today announced its consolidated operating results for the fourth quarter of 2022.

Fourth quarter consolidated revenue was NT$67.84 billion, decreasing 10.0% QoQ from NT$75.39 billion in 3Q22. Compared to a year ago, 4Q22 revenue grew 14.8% YoY from NT$59.10 billion in 4Q21. Consolidated gross margin for 4Q22 was 42.9%. Net income attributable to the shareholders of the parent was NT$19.1 billion, with earnings per ordinary share of NT$1.54.

Jason Wang, co-president of UMC, said, "In the fourth quarter, due to a significant slowdown across most of our end markets and inventory correction in the semiconductor industry, our wafer shipments fell 14.8% QoQ while overall fab utilization rate dropped to 90%. Average selling price increased slightly during the quarter as a result of our ongoing product mix optimization efforts, moderating the decline in revenue."

[1]Unless otherwise stated, all financial figures discussed in this announcement are prepared in accordance with TIFRSs recognized by Financial Supervisory Commission in the ROC, which is different from IFRSs issued by the International Accounting Standards Board. They represent comparisons among the three-month period ending December 31, 2022, the three-month period ending September 30, 2022, and the equivalent three-month period that ended December 31, 2021. For all 4Q22 results, New Taiwan Dollar (NT$) amounts have been converted into U.S. Dollars at the December 31, 2022 exchange rate of NT$ 30.70 per U.S. Dollar.

"For the full year 2022, UMC's revenue hit a record high of NT$278.7 billion while operating income exceeded NT$100 billion. Gross margin reached 45%, driven by a more favorable foreign exchange rate, expanding 22/28nm portfolio, and newly added capacity. We had taken advantage of the industry upturn over the past two years to enhance our differentiation in specialty technology offering, improve profitability, and deepen relationships with key customers. Revenue from 22/28nm technologies increased more than 56% YoY, driven by our industry-leading 28nm process for OLED display drivers and image signal processors. Our automotive segment also delivered impressive growth in 2022, increasing 82% YoY to account for approximately 9% of total sales. We expect this segment will continue to be a key growth catalyst in 2023 and beyond, driven by the long-term trend of vehicle electrification and automation. UMC is well positioned to serve the market with our comprehensive portfolio of auto-grade process technologies and facilities certified according to rigorous quality standards, while we continue to build strong partnerships with world-class automotive leaders."

Co-president Wang commented, "Given the soft global economic outlook for 2023, we expect the current challenging environment to persist through the first quarter as customers' days of inventory are still higher than normal while order visibility remains low. To manage this period of weakness, the Company is implementing strict cost control measures and deferring certain capital expenditures where possible. In the longer term, we remain positive that UMC's differentiated specialty technology leadership, geographically diversified capacity offering, and quality and operational excellence will enable the Company to capture demand fueled by continuous digital transformation across industries and be the foundry of choice for leading customers."

Co-president Wang added, "In 2022, we took solid steps in executing our net zero by 2050 roadmap. As part of our efforts to reduce emissions across the entire value chain, UMC recently introduced a program to empower our suppliers with resources to measure and manage their emissions output. To round out the important progress we made towards our ESG goals this year, we were honored to receive recognition from domestic and international institutions. In the 2022 Dow Jones Sustainability Indices (DJSI), UMC was ranked first in terms of overall sustainability performance among semiconductor foundry peers in the 2022 DJSI, while we were the only semiconductor firm globally to achieve double-A scoring for climate change and water security in CDP's annual evaluation of corporate environmental action. Sustainability and a company's long-term success are inextricably linked, and UMC will continue to strive to meet expectations of all of our stakeholders while acting as responsible social and environment stewards."

Summary of Operating Results

Operating Results					
(Amount: NT$ million)	4Q22	3Q22	QoQ % change	4Q21	YoY % change
Operating Revenues	67,836	75,392	(10.0)	59,100	14.8
Gross Profit	29,124	35,664	(18.3)	23,103	26.1
Operating Expenses	(6,798)	(6,794)	0.1	(6,821)	(0.3)
Net Other Operating Income and Expenses	1,311	1,287	1.9	1,334	(1.7)
Operating Income	23,637	30,157	(21.6)	17,616	34.2
Net Non-Operating Income and Expenses	889	2,189	(59.4)	558	59.4
Net Income Attributable to Shareholders of the Parent	19,068	26,996	(29.4)	15,949	19.6
EPS (NT$ per share)	1.54	2.19		1.30	
(US$ per ADS)	0.251	0.357		0.212	

Fourth quarter operating revenues declined by 10.0% sequentially to NT$67.84 billion resulting from the inventory correction within the semi industry which lowered wafer shipments. Revenue contribution from 40nm and below technologies represented 45% of wafer revenue. Gross profit decreased 18.3% QoQ to NT$29.12 billion, or 42.9% of revenue. Operating expenses remained flat at NT$6.80 billion. Net other operating income increased to NT$1.31 billion. Net non-operating income totaled NT$0.89 billion. Net income attributable to shareholders of the parent amounted to NT$19.07 billion.

Earnings per ordinary share for the quarter was NT$1.54. Earnings per ADS was US$0.251. The basic weighted average number of shares outstanding in 4Q22 was 12,348,880,384, compared with 12,305,516,644 shares in 3Q22 and 12,254,114,875 shares in 4Q21. The diluted weighted average number of shares outstanding was 12,684,106,050 in 4Q22, compared with 12,635,661,561 shares in 3Q22 and 12,489,949,678 shares in 4Q21. The fully diluted shares counted on December 31, 2022 were approximately 12,684,106,000.

Detailed Financials Section

Operating revenues decreased to NT$67.84 billion. COGS declined 2.6% to NT$38.71 billion, which included 7.5% sequential decrease in depreciation. Gross profit fell 18.3% QoQ to NT$29.12 billion. Operating expenses remained flat at NT$6.80 billion, as Sales & Marketing reduced 10.2% to NT$0.95 billion while R&D was up 3.1% QoQ to NT$3.41 billion, representing 5.0% of revenue. Net other operating income was NT$1.31 billion. In 4Q22, operating income declined 21.6% QoQ to NT$23.64 billion.

COGS & Expenses

(Amount: NT$ million)	4Q22	3Q22	QoQ % change	4Q21	YoY % change
Operating Revenues	67,836	75,392	(10.0)	59,100	14.8
COGS	(38,712)	(39,728)	(2.6)	(35,997)	7.5
Depreciation	(8,898)	(9,622)	(7.5)	(10,122)	(12.1)
Other Mfg. Costs	(29,814)	(30,106)	(1.0)	(25,875)	15.2
Gross Profit	29,124	35,664	(18.3)	23,103	26.1
Gross Margin (%)	42.9%	47.3%		39.1%	
Operating Expenses	(6,798)	(6,794)	0.1	(6,821)	(0.3)
G&A	(2,438)	(2,428)	0.4	(2,164)	12.7
Sales & Marketing	(953)	(1,061)	(10.2)	(1,240)	(23.1)
R&D	(3,407)	(3,304)	3.1	(3,414)	(0.2)
Expected Credit Impairment Loss	(0)	(1)	(48.5)	(3)	(88.3)
Net Other Operating Income & Expenses	1,311	1,287	1.9	1,334	(1.7)
Operating Income	23,637	30,157	(21.6)	17,616	34.2

Net non-operating income in 4Q22 was NT$0.89 billion, primarily reflecting the NT$1.38 billion in net investment gain and the NT$0.58 billion in net interest income partially offset by the NT$1.06 billion in exchange loss.

Non-Operating Income and Expenses

(Amount: NT$ million)	4Q22	3Q22	4Q21
Non-Operating Income and Expenses	889	2,189	558
Net Interest Income and Expenses	584	139	(343)
Net Investment Gain and Loss	1,382	780	2,689
Exchange Gain and Loss	(1,057)	1,293	312
Other Gain and Loss	(20)	(23)	(2,100)

In 4Q22, cash inflow from operating activities was NT$40.96 billion. Cash outflow from investing activities amounted to NT$36.05 billion, which included NT$36.32 billion in capital expenditure, resulting in free cash flow of NT$4.63 billion. Cash outflow from financing was NT$9.19 billion, primarily from a NT$8.08 billion in bank loans and a NT$2.54 billion in redemption of bonds. Net cash outflow in 4Q22 totaled NT$6.83 billion. Over the next 12 months, the company expects to repay NT$ 2.49 billion in bank loans.

`

Cash Flow Summary

(Amount: NT$ million)	For the 3-Month Period Ended Dec. 31, 2022	For the 3-Month Period Ended Sep.30, 2022
Cash Flow from Operating Activities	40,956	39,696
Net income before tax	24,526	32,346
Depreciation & Amortization	10,477	11,195
Share of profit of associates and joint ventures	(1,705)	(328)
Income tax paid	(385)	(652)
Changes in working capital & others	8,043	(2,865)
Cash Flow from Investing Activities	(36,045)	(21,419)
Decrease in financial assets measured at amortized cost	9	2,070
Acquisition of PP&E	(35,951)	(22,466)
Acquisition of intangible assets	(440)	(1,080)
Others	337	57
Cash Flow from Financing Activities	(9,185)	(26,688)
Bank loans	(8,082)	(473)
Redemption of bonds	(2,542)	(1,031)
Increase (decrease) in deposits-in	(389)	14,441
Cash distributed from additional paid-in capital	-	(37,446)
Others	1,828	(2,179)
Effect of Exchange Rate	(2,556)	5,337
Net Cash Flow	(6,830)	(3,074)
Beginning balance	180,649	183,723
Ending balance	173,819	180,649

Cash and cash equivalents slightly decreased to NT$173.82 billion. Days of inventory increased by 6 days to 72 days.

Current Assets			
(Amount: NT$ billion)	4Q22	3Q22	4Q21
Cash and Cash Equivalents	173.82	180.65	132.62
Accounts Receivable	36.98	44.84	35.19
Days Sales Outstanding	55	53	53
Inventories, net	31.07	30.10	23.01
Days of Inventory	72	66	59
Total Current Assets	252.37	266.95	233.27

Current liabilities slightly increased to NT$108.57 billion. Long-term credit/bonds decreased to NT$39.88 billion. Total liabilities decreased to NT$197.60 billion, leading to a debt to equity ratio of 59%.

Liabilities			
(Amount: NT$ billion)	4Q22	3Q22	4Q21
Total Current Liabilities	108.57	108.01	105.45
Accounts Payable	8.98	10.04	8.36
Short-Term Credit / Bonds	7.59	12.19	39.26
Payables on Equipment	18.63	19.40	7.88
Other	73.37	66.38	49.95
Long-Term Credit / Bonds	39.88	46.15	39.83
Long-Term Investment Liabilities	4.22	4.26	8.25
Total Liabilities	197.60	204.21	183.22
Debt to Equity	59%	65%	65%

Analysis of Revenue[2]

Revenue from Asia-Pacific declined to 54% while business from North America was 30% of sales. Business from Europe was 9% while contribution from Japan increased to 7%.

Revenue Breakdown by Region

Region	4Q22	3Q22	2Q22	1Q22	4Q21
North America	30%	23%	22%	22%	21%
Asia Pacific	54%	62%	65%	64%	66%
Europe	9%	9%	8%	8%	7%
Japan	7%	6%	5%	6%	6%

Revenue contribution from 22/28nm grew to 28% of the wafer revenue, while 40nm contribution was 17% of sales.

Revenue Breakdown by Geometry

Geometry	4Q22	3Q22	2Q22	1Q22	4Q21
14nm and below	0%	0%	0%	0%	0%
14nm<x<=28nm	28%	25%	22%	20%	20%
28nm<x<=40nm	17%	17%	18%	18%	18%
40nm<x<=65nm	17%	18%	19%	19%	19%
65nm<x<=90nm	9%	8%	7%	8%	8%
90nm<x<=0.13um	12%	12%	12%	12%	12%
0.13um<x<=0.18um	10%	10%	12%	13%	13%
0.18um<x<=0.35um	5%	8%	8%	7%	7%
0.5um and above	2%	2%	2%	3%	3%

Revenue from fabless customers accounted for 81% of revenue.

Revenue Breakdown by Customer Type

Customer Type	4Q22	3Q22	2Q22	1Q22	4Q21
Fabless	81%	83%	86%	87%	86%
IDM	19%	17%	14%	13%	14%

Revenue from the communication segment represented 45%, while business from computer applications decreased to 12%. Business from consumer applications was 25% as other segments increased to 18% of revenue.

Revenue Breakdown by Application (1)

Application	4Q22	3Q22	2Q22	1Q22	4Q21
Computer	12%	14%	16%	17%	17%
Communication	45%	45%	45%	45%	46%
Consumer	25%	27%	27%	26%	26%
Others	18%	14%	12%	12%	11%

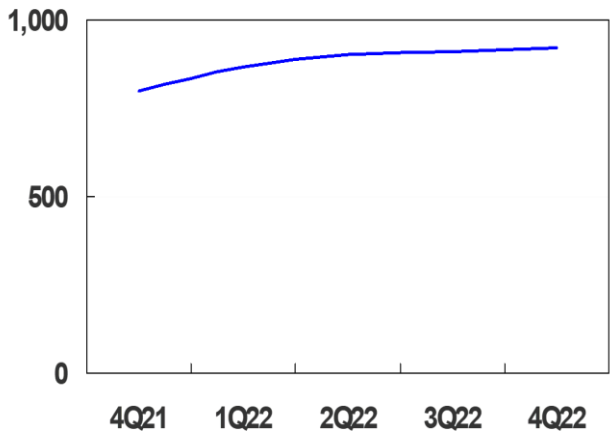
(1) ***Computer*** consists of ICs such as CPU, GPU, HDD controllers, DVD/CD-RW control ICs, PC chipset, audio codec, keyboard controller, monitor scaler, USB, I/O chipset, WLAN. ***Communication*** consists of handset components, broadband, bluetooth, Ethernet, LAN, DSP, etc. ***Consumer*** consists of ICs used for DVD players, DTV, STB, MP3/MP4, flash controller, game consoles, DSC, smart cards, toys, etc.

[2] Revenue in this section represents wafer sales

Blended ASP Trend

Blended average selling price (ASP) grew slightly in 4Q22.

Unit: USD



Shipment and Utilization Rate[3]

Wafer shipments decreased 14.8% QoQ to 2,213K in the fourth quarter, while quarterly capacity was 2,543K. Overall utilization rate in 4Q22 declined to 90%.

Wafer Shipments

	4Q22	3Q22	2Q22	1Q22	4Q21
Wafer Shipments (8" K equivalents)	2,213	2,597	2,622	2,513	2,546

Quarterly Capacity Utilization Rate

	4Q22	3Q22	2Q22	1Q22	4Q21
Utilization Rate	90%	100%+	100%+	100%+	100%+
Total Capacity (8" K equivalents)	2,543	2,539	2,528	2,420	2,419

[3] Utilization Rate = Quarterly Wafer Out / Quarterly Capacity

Capacity[4]

Total capacity in the fourth quarter increased to 2,543K 8-inch equivalent wafers. Capacity will decline in the first quarter of 2023 to 2,522K 8-inch equivalent wafers, reflecting the annual production maintenance across 8" and 12" facilities.

Annual Capacity in *thousands of wafers*

FAB		Geometry (um)	2022	2021	2020	2019
WTK	6"	5 – 0.15	335	329	371	370
8A	8"	3 – 0.11	765	755	802	825
8C	8"	0.35 – 0.11	459	459	452	436
8D	8"	0.18 – 0.09	410	380	371	359
8E	8"	0.6 – 0.14	469	457	449	426
8F	8"	0.18 – 0.11	550	514	485	434
8S	8"	0.18 – 0.11	443	408	373	372
8N	8"	0.5 – 0.11	952	917	917	831
12A	12"	0.13 – 0.014	1,170	1,070	1,044	997
12i	12"	0.13 – 0.040	655	641	628	595
12X	12"	0.080 – 0.022	314	284	217	203
12M	12"	0.13 – 0.040	436	395	391	98
Total[(1)]			10,031	9,453	9,188	8,148
YoY Growth Rate			6%	3%	13%	6%

Quarterly Capacity in *thousands of wafers*

FAB		1Q23E	4Q22	3Q22	2Q22
WTK	6"	80	85	85	84
8A	8"	189	192	192	192
8C	8"	113	115	115	115
8D	8"	101	103	103	103
8E	8"	116	118	118	118
8F	8"	136	138	138	138
8S	8"	109	111	111	111
8N	8"	244	245	242	235
12A	12"	305	301	301	301
12i	12"	162	164	164	164
12X	12"	78	80	80	78
12M	12"	108	110	110	110
Total		**2,522**	**2,543**	**2,539**	**2,528**

(1) One 6-inch wafer is converted into 0.5625 ($6^2/8^2$) 8-inch equivalent wafer; one 12-inch wafer is converted into 2.25 ($12^2/8^2$) 8-inch equivalent wafers. Total capacity figures are expressed in 8-inch equivalent wafers.

CAPEX

CAPEX spending in 4Q22 totaled US$1,167 million as 2022 CAPEX amounted to US$2.7 billion. 2023 cash-based CAPEX budget will be US$3.0 billion.

Capital Expenditure by Year - in US$ billion

Year	2022	2021	2020	2019	2018
CAPEX	$ 2.7	$ 1.8	$ 1.0	$ 0.6	$ 0.7

2023 CAPEX Plan

8"	12"	Total
10%	90%	US$3.0 billion

[4] Estimated capacity numbers are based on *calculated maximum output* rather than *designed capacity*. The actual capacity numbers may differ depending upon equipment delivery schedules, pace of migration to more advanced process technologies, and other factors affecting production ramp-up.

Brief Summary of Full Year 2022 Consolidated Results

- Consolidated revenue in NTD increased 30.8% YoY to NT$278.71 billion, from NT$213.01 billion in 2021.
- Gross profit increased 74.6%, compared to a year ago, representing 45.1% of 2022 revenue.
- Operating income grew 101.8% year on year, accounting for 37.4% of 2022 revenue.
- Net income attributable to shareholders of the parent increased 56.3% to NT$87.20 billion in 2022.
- EPS was NT$7.09, or EPADS of US$1.155 for 2022.
- 22/28nm revenue grew 56% compared to 2021, as contribution from 28nm technologies and below accounted for 24% in 2022.

Operating Results			
(Amount: NT$ million)	2022	2021	YoY % change
Operating Revenues	278,705	213,011	30.8
Gross Profit	125,764	72,050	74.6
Operating Expenses	(26,812)	(25,590)	4.8
Net Other Operating Income and Expenses	5,340	5,226	2.2
Operating Income	104,292	51,686	101.8
Net Non-Operating Income and Expenses	1,805	10,117	(82.2)
Income Tax Expense	(18,079)	(6,691)	170.2
Net Income Attributable to Shareholders of the Parent	87,198	55,780	56.3
EPS (NT$ per share)	7.09	4.57	
(US$ per ADS)	1.155	0.744	

Annual Sales Breakdown in Revenue for Foundry Segment

Region	2022	2021
North America	24%	22%
Asia Pacific	61%	65%
Europe	9%	7%
Japan	6%	6%

Technology	2022	2021
14nm and below	0%	0%
14nm<x<=28nm	24%	20%
28nm<x<=40nm	18%	18%
40nm<x<=65nm	18%	19%
65nm<x<=90nm	8%	8%
90nm<x<=0.13um	12%	12%
0.13um<x<=0.18um	11%	13%
0.18um<x<=0.35um	7%	8%
0.5um and above	2%	2%

Customer Type	2022	2021
Fabless	84%	85%
IDM	16%	15%

Application	2022	2021
Computer	15%	17%
Communication	45%	46%
Consumer	26%	27%
Others	14%	10%

First Quarter 2023 Outlook & Guidance

Quarter-over-Quarter Guidance:

- Wafer Shipments: To decrease in the high teens % range
- ASP in USD: To remain flat
- Gross Profit Margin: Will be in the mid-30% range
- Capacity Utilization: approximately 70%
- 2023 CAPEX: US$3.0 billion

Recent Developments / Announcements

Nov. 9, 2022	UMC Drives Low-carbon Supply Chain with GHG Inventory Initiative
Nov. 30, 2022	Cadence and UMC Certified mmWave Reference Flow Delivers First-Pass Silicon Success
Dec. 10, 2022	UMC Again Leads Semiconductor Foundries in the 2022 Dow Jones Sustainability Indices
Dec. 16, 2022	UMC Earns Top Rating in CDP's Climate Change and Water Security Assessments
Dec. 30, 2022	UMC Achieves AA Certification for Excellence in Intellectual Property Management and Protection

Please visit UMC's website for further details regarding the above announcements

Conference Call / Webcast Announcement

Monday, January 16, 2023

Time: 5:00 PM (Taipei) / 4:00 AM (New York) / 09:00 AM (London)

Dial-in numbers and Access Codes:

Hong Kong Number:	+852-2112-1888
Taiwan Number:	02-2162-6306
Other Areas:	+852-2112-1888
Access Code:	9603516#

A live webcast and replay of the 4Q22 results announcement will be available at www.umc.com under the “Investors / Events” section.

About UMC

UMC (NYSE: UMC, TWSE: 2303) is a leading global semiconductor foundry. The company provides high quality IC production with a focus on both logic and specialty technologies to serve every major sector of the electronics industry. UMC’s comprehensive technology and manufacturing solutions include logic/RF, embedded high voltage, embedded flash, RFSOI/BCD and IATF-16949 automotive manufacturing certification for all its manufacturing facilities. UMC operates 12 fabs that are strategically located throughout Asia with a maximum capacity of approximately 850,000 8-inch equivalent wafers per month. The company employs approximately 20,000 people worldwide, with offices in Taiwan, China, United States, Europe, Japan, Korea and Singapore. For more information, please visit: http://www.umc.com.

Safe Harbor Statements

This press release contains forward-looking statements within the meaning of Section 27A of the United States Securities Act of 1933, as amended, and Section 21E of the United States Securities Exchange Act of 1934, as amended, and as defined in the United States Private Securities Litigation Reform Act of 1995. These forward-looking statements include, but are not limited to, statements regarding anticipated financial results for the fourth quarter of 2022; the expected wafer shipment and ASP; the anticipated annual budget; capex strategies; environmental protection goals and water management strategies; impact of foreign currency exchange rates; expected foundry capacities; the ability to obtain new business opportunities; and information under the heading "Fourth Quarter of 2022 Outlook and Guidance."

These forward-looking statements involve known and unknown risks, uncertainties and other factors that may cause the actual performance, financial condition or results of operations of UMC to be materially different from what is stated or may be implied in such forward-looking statements. Investors are cautioned that actual events and results could differ materially from those statements as a result of a number of factors including, but not limited to: (i) dependence upon the frequent introduction of new services and technologies based on the latest developments in the industry in which UMC operates; (ii) the intensely competitive semiconductor, communications, consumer electronics and computer industries and markets; (iii) the risks associated with international business activities; (iv) dependence upon key personnel; (v) general economic and political conditions; (vi) possible disruptions in commercial activities caused by natural and human-induced events and disasters, including natural disasters, terrorist activity, armed conflict and highly contagious diseases; (vii) reduced end-user purchases relative to expectations and orders; and (viii) fluctuations in foreign currency exchange rates. Further information regarding these and other risk factors is included in UMC's filings with the United States Securities and Exchange Commission, including its Annual Report on Form 20-F. All information provided in this release is as of the date of this release and are based on assumptions that UMC believes to be reasonable as of this date, and UMC does not undertake any obligation to update any forward-looking statement as a result of new information, future events or otherwise, except as required under applicable law.

The financial statements included in this release are prepared and published in accordance with Taiwan International Financial Reporting Standards, or TIFRSs, recognized by the Financial Supervisory Commission in the ROC, which is different from International Financial Reporting Standards, or IFRSs, issued by the International Accounting Standards Board. Investors are cautioned that there may be significant differences between TIFRSs and IFRSs. In addition, TIFRSs and IFRSs differ in certain significant respects from generally accepted accounting principles in the ROC and generally accepted accounting principles in the United States.

- FINANCIAL TABLES TO FOLLOW -